

Cyclic extensions of elliptic curves of j -invariant 1728

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Abstract

In [6] the curve $E_1 : y^2 = x^3 + x$ was treated in detail: for a split Gaussian prime π with $N(\pi) = p \equiv 1 \pmod{4}$, the division polynomial ψ_π was shown to be Eisenstein at π , and the coordinates of a single π -torsion point were shown to generate a cyclic extension $K(x_1, y_1)/K$ of degree $N(\pi) - 1$ over $K = \mathbb{Q}(i)$, with Galois group $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$. Here we ask how this picture changes when the coefficient 1 is replaced by an arbitrary integer: $E_a : y^2 = x^3 + ax$, $a \in \mathbb{Z} \setminus \{0\}$. We show first that the entire derivation is *uniform* in a : for every a with $\pi \nmid 2a$, the same cyclic group of order $N(\pi) - 1$ arises, the only modification to the argument being a weighted-homogeneity computation pinning down the exact dependence of the constant term $\psi_\pi(0) = u_\pi a^{(N(\pi)-1)/4}$ on a , where $u_\pi \in \mathbb{Z}[i]^\times$ is the unit computed in [6]. So the *abstract* Galois group never depends on a . What does depend on a is the actual field $K(x_1, y_1) \subset \overline{K}$: we construct an explicit isomorphism $E_1 \rightarrow E_a$, defined over $K(\sqrt[4]{a})$, that intertwines the $\mathbb{Z}[i]$ -actions, and use it to show that $K_a(\pi) := K(x_1, y_1)$ for E_a and $K_1(\pi)$ for E_1 agree on a specific subfield of index $d = [K(\sqrt[4]{a}) : K] \in \{1, 2, 4\}$ – a quantity depending only on the class of a in $\mathbb{Q}(i)^\times/(\mathbb{Q}(i)^\times)^4$, not on π – and, under a mild disjointness hypothesis, generate together a field of degree $d(N(\pi) - 1)$ over K . We verify this structure completely explicitly, with no hypothesis needed, for the smallest split prime $\pi = 1 + 2i$ ($N(\pi) = 5$), where an elementary computation gives $\psi_{1+2i}^{(a)}(x) = (1 + 2i)x^2 + a$ for every a , and we exhibit $a = 16$, $a = 4$, $a = 2$ realizing the three cases $d = 1, 2, 4$ by hand.

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1 Introduction

Fix $K = \mathbb{Q}(i)$ and consider the one-parameter family of elliptic curves

$$E_a : y^2 = x^3 + ax, \quad a \in \mathbb{Z} \setminus \{0\}.$$

Every member of this family has j -invariant 1728, and every member admits the same complex multiplication:

$$[i] : (x, y) \mapsto (-x, iy)$$

is an automorphism of E_a for *every* a , since $(-x)^3 + a(-x) = -(x^3 + ax)$ and $(iy)^2 = -y^2$. Thus $\text{End}(E_a) \cong \mathbb{Z}[i]$ and $\text{Aut}(E_a) \cong \mathbb{Z}[i]^\times = \{\pm 1, \pm i\}$ regardless of a . This is exactly the situation studied for $a = 1$ in [6], where the main theorem produced, for each split Gaussian prime π (i.e. $N(\pi) = p \equiv 1 \pmod{4}$) with $\pi \nmid 2$, a cyclic extension $K(x_1, y_1)/K$ of degree $N(\pi) - 1$, Galois group $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$, realized by the coordinates of a single nonzero π -torsion point $P_1 = (x_1, y_1) \in E_1$.

The present note asks the natural next question: *what role does the coefficient a actually play in this derivation?* We give a complete answer in two parts.

- (1) **The abstract group is a -independent.** We show in §§2–3 that the whole argument of [6, Ch. 3–4] – the decomposition of $E_a[p]$ into $[i]$ -eigenspaces, the factorization $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$, the Eisenstein criterion, the degree computations, and the identification of $\text{Gal}(K(x_1, y_1)/K)$ with $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ – goes through *verbatim* for every $a \in \mathbb{Z} \setminus \{0\}$ with $\pi \nmid 2a$. The only place where the argument of [6] must actually be modified is the claim that $\psi_\pi(0)$ is a unit: this is special to $a = 1$. We replace it with the correct general statement, a weighted-homogeneity computation showing

$$\psi_\pi(0) = u_\pi \cdot a^{(N(\pi)-1)/4}, \quad u_\pi \in \mathbb{Z}[i]^\times,$$

and observe that the Eisenstein argument only ever needed the weaker fact $\pi \nmid \psi_\pi(0)$, which follows immediately from $\pi \nmid a$.

- (2) **The concrete field is a quartic twist.** Although the *isomorphism type* of the Galois group is fixed, the actual field $K(x_1, y_1) \subset \bar{K}$ genuinely depends on a . We make this precise in §§4–5 by constructing the scaling isomorphism $\phi_t : E_1 \rightarrow E_a$, $(x, y) \mapsto (t^2x, t^3y)$ with $t^4 = a$, defined over $K(t) = K(\sqrt[4]{a})$, and showing it intertwines the $\mathbb{Z}[i]$ -module structures on π -torsion. This lets us describe exactly how $K_a(\pi) :=$

$K(x_1, y_1)$ for E_a sits relative to $K_1(\pi)$ for E_1 : the two fields agree on a specific subfield of index $d := [K(\sqrt[4]{a}) : K] \in \{1, 2, 4\}$, a quantity that depends only on the class of a in $K^\times / (K^\times)^4$ and *not* on π at all.

Finally, in §6, we carry this out with no appeal to any general hypothesis, entirely by hand, for the smallest split prime $\pi = 1 + 2i$ ($N(\pi) = 5$), reproducing and extending the worked example of [6, Ch. 4].

2 Preliminaries

We record facts about $\mathbb{Z}[i]$ used freely below; see [1, Ch. 1–2] or [2, Ch. 9].

- $\mathbb{Z}[i]$ is a PID with norm $N(n + im) = n^2 + m^2$ and unit group $\mathbb{Z}[i]^\times = \{1, i, -1, -i\}$, cyclic of order 4.
- An odd rational prime p splits as $p = \pi\bar{\pi}$ in $\mathbb{Z}[i]$ iff $p \equiv 1 \pmod{4}$; it is inert iff $p \equiv 3 \pmod{4}$. In the split case $4 \mid p - 1$, so $\mathbb{Z}[i]^\times$ embeds canonically into the cyclic group $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ of order $p - 1$ as its unique subgroup of order 4.
- Throughout, $\pi = n + im$ denotes a split Gaussian prime with $N(\pi) = p \equiv 1 \pmod{4}$, $\pi \nmid 2$.

3 Uniformity of the Main Theorem in a

Fix $a \in \mathbb{Z} \setminus \{0\}$ and write $E = E_a : y^2 = x^3 + ax$. Its discriminant is $\Delta_a = -64a^3$, so E has good reduction at every odd prime not dividing a .

3.1 The torsion module and the composite polynomial

The proof in [6, Ch. 3, “Structure of the Gaussian Torsion Subgroup”] that $E_1[p] \cong E_1[\pi] \oplus E_1[\bar{\pi}]$ for $p \equiv 1 \pmod{4}$ (and that $E_1[p]$ is simple for $p \equiv 3 \pmod{4}$) uses only the relation $[i]^2 = [-1]$ on the $\mathbb{F}_p$ -vector space $E[p]$ and elementary eigenvalue counting for the characteristic polynomial $\varphi^2 + 1$. Nothing in that argument refers to a : $[i]^2 = [-1]$ holds on every E_a . Hence:

Theorem 3.1 (Structure of the torsion module, any a). *For $p \nmid 2a$: if $p \equiv 1 \pmod{4}$ then $E_a[p] \cong E_a[\pi] \oplus E_a[\bar{\pi}]$ for $p = \pi\bar{\pi}$; if $p \equiv 3 \pmod{4}$ then $E_a[p]$ is a simple $\mathbb{Z}[i]$ -module.*

Let $\psi_n, \phi_n, \omega_n \in \mathbb{Z}[a][x, y]$ denote the standard division polynomials for $y^2 = x^3 + ax$ (i.e. the general recursions of [5, Ch. III] or [7, Lemma 3.4–3.5] specialized to $b = 0$), and let

$$\Psi_\pi := \phi_m \psi_n^2 + \phi_n \psi_m^2 \in \mathbb{Z}[a][x]$$

be the composite polynomial of [6, Ch. 3, §“Derivation of Ψ_π and Φ_π ”], constructed from $[\pi]P = [n]P + [i]([m]P)$ exactly as there; this derivation is again purely formal in a (it only uses the addition law and the formula $[i](x, y) = (-x, iy)$, both a -independent). Theorem 3.1 gives the factorization exactly as in [6, Ch. 3, “Structure of the Gaussian Torsion Subgroup” and the theorem following it]:

Theorem 3.2. $\Psi_\pi = \psi_\pi \psi_{\bar{\pi}}$ in $\mathbb{Z}[i][a][x]$, where ψ_π is the polynomial vanishing exactly on the x -coordinates of $E_a[\pi] \setminus \{\mathcal{O}\}$.

3.2 Weighted homogeneity and the constant term

The one place [6, Ch. 3] used the specific value $a = 1$ was in asserting that $\psi_\pi(0)$ is always a unit (their corollary on the trailing coefficient, used in the constant-term step of the Eisenstein argument). This is false for general a – e.g. we compute below that $\psi_{1+2i}(0) = a$ exactly – but it is also *more than what the Eisenstein argument needs*. We isolate the correct general fact.

Lemma 3.3 (Weighted homogeneity). *Assign weights $\text{wt}(x) = 2$, $\text{wt}(y) = 3$, $\text{wt}(a) = 4$. Then $\psi_n(x, y; a)$ is weighted-homogeneous of weight $n^2 - 1$, i.e.*

$$\psi_n(\lambda^2 x, \lambda^3 y; \lambda^4 a) = \lambda^{n^2-1} \psi_n(x, y; a)$$

for every λ . Consequently ϕ_n has weight $2n^2$, and ψ_π (for $\pi = n + im$) is weighted-homogeneous of weight $N(\pi) - 1$.

Proof. This is immediate by induction from the standard recursions $\psi_0 = 0, \psi_1 = 1, \psi_2 = 2y, \psi_3 = 3x^4 + 6ax^2 - a^2, \psi_4 = 4y(x^6 + 5ax^4 - 5a^2x^2 - a^3)$ (setting $b = 0$ in [7, Lemma 3.4]), and

$$\psi_{2n+1} = \psi_{n+2}\psi_n^3 - \psi_{n-1}\psi_{n+1}^3, \quad \psi_{2n} = \frac{\psi_n}{2y}(\psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2) :$$

every monomial occurring in these formulas has total weight equal to the claimed weight of the left-hand side, using $y^2 = x^3 + ax$ (weight 6 on both sides) to eliminate any residual y 's. The claim for $\phi_n = x\psi_n^2 - \psi_{n+1}\psi_{n-1}$ follows immediately, since both terms have weight $2n^2$. For ψ_π : $\Psi_\pi = \phi_m\psi_n^2 + \phi_n\psi_m^2$ has weight $2m^2 + 2(n^2 - 1) = 2n^2 + 2(m^2 - 1) = 2(n^2 + m^2) - 2 = 2N(\pi) - 2$; since $\Psi_\pi = \psi_\pi\psi_{\bar{\pi}}$ with $\deg_x \psi_\pi = \deg_x \psi_{\bar{\pi}} = \frac{1}{2}(N(\pi) - 1)$ (the same leading-order degree computation as [6, Ch. 4, degree lemma preceding the Galois-group theorem], which never used $a = 1$), and complex conjugation preserves weighted-homogeneity, each factor inherits weight $N(\pi) - 1$. $\square$

Corollary 3.4 (The constant term). $\psi_\pi(0; a) = u_\pi \cdot a^{(N(\pi)-1)/4}$, where $u_\pi := \psi_\pi(0; 1) \in \mathbb{Z}[i]^\times$ is the unit shown in [6, Ch. 3, trailing-coefficient corollary] to occur as the constant term of ψ_π for E_1 . In particular $\pi \mid \psi_\pi(0; a)$ if and only if $\pi \mid a$.

Proof. Setting $x = 0$ in Lemma 3.3 with $\lambda = a^{1/4}$ (formally) gives $\psi_\pi(0; a) = a^{(N(\pi)-1)/4} \psi_\pi(0; 1)$; note $4 \mid N(\pi) - 1 = p - 1$ since $p \equiv 1 \pmod{4}$, so the exponent is a genuine nonnegative integer, and the identity of polynomials in a obtained this way is literal (both sides are, by Lemma 3.3, single monomials in a up to the fixed factor $\psi_\pi(0; 1)$). Since u_π is a unit, $\pi \mid u_\pi a^{(p-1)/4}$ iff $\pi \mid a$. $\square$

3.3 The generalized main theorem

Theorem 3.5 (Eisenstein irreducibility, any a). *Let $\pi \nmid 2a$. Then $\psi_\pi(x; a)$ is irreducible over $K = \mathbb{Q}(i)$.*

Proof. We check the Eisenstein criterion in the reversed form used in [6, Ch. 3, main theorem of the chapter]: (i) the leading coefficient of ψ_π is π up to a unit, independent of a (it comes from the n^2, m^2 -leading terms of $\phi_n \psi_m^2, \phi_m \psi_n^2$, which are a -independent by the standard leading-term computation, e.g. [7, Lemma 3.5]), so $\pi \mid$ leading coefficient but $\pi^2 \nmid$ it; (ii) the reduction $\tilde{E}_a$ of E_a modulo π is well-defined (good reduction, since $\pi \nmid \Delta_a = -64a^3$) and the reduced endomorphism $\widetilde{[\pi]}$ is the p -power Frobenius [4, Ch. II, Prop. 5.3], purely inseparable of degree p , forcing every non-leading, non-constant coefficient of ψ_π to vanish mod π , exactly as in [6, Ch. 3]; (iii) by Corollary 3.4, $\pi \nmid \psi_\pi(0; a)$ since $\pi \nmid a$. This is exactly the hypothesis needed; we do not need $\psi_\pi(0; a)$ to be a unit. $\square$

Theorem 3.6 (Generalized main theorem). *Let $\pi \nmid 2a$, $N(\pi) = p$, and let $P_1 = (x_1, y_1)$ be a nonzero π -torsion point on E_a . Then:*

1. $[K(x_1) : K] = \frac{1}{2}(p-1)$, and $K(x_1)/K$ is cyclic Galois, $\text{Gal}(K(x_1)/K) \cong (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times / \{\pm 1\}$;
2. $K(y_1^2) = K(x_1)$;
3. $[K(x_1, y_1) : K] = p - 1$, and $K(x_1, y_1)/K$ is cyclic Galois with $\text{Gal}(K(x_1, y_1)/K) \cong (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$.

Proof. Part (1) is Theorem 3.5 together with the degree count $\deg \psi_\pi = \frac{1}{2}(N(\pi) - 1)$ (Lemma 3.3 plus Theorem 3.2, unchanged from [6, Ch. 4, degree lemma preceding the Galois-group theorem]) and the same Galois-group argument as [6, Ch. 4, “Galois action on the x -coordinate field” theorem] ($\sigma(x_1) = x([\alpha]P_1)$ for $\alpha \in (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$, kernel $\{\pm 1\}$ from $x(P) = x(-P)$).

For (2): $y_1^2 = x_1^3 + ax_1 = x_1(x_1^2 + a)$. Since $a \in K \subseteq K(x_1)$, clearly $K(y_1^2) \subseteq K(x_1)$. For the reverse inclusion, the argument of [6, Ch. 4, proposition bridging the x_1 and y_1 extensions] applies verbatim with $\alpha := x_1^3 + ax_1$: any Galois conjugate x_k of x_1 over $K(\alpha)$ satisfies $x_k^3 + ax_k = \alpha = x_1^3 + ax_1$, hence $y_k^2 = y_1^2$, hence $P_k = \pm P_1$, hence $x_k = x_1$; the coefficient a never enters this chain of implications beyond being present in $K(\alpha)$. So $[K(x_1) : K(\alpha)] = 1$ and $K(x_1) = K(\alpha) = K(y_1^2)$.

For (3): exactly as in [6, Ch. 4, the proposition and theorem on the full extension $K(y_1)$]: since ψ_π is Eisenstein at π (Theorem 3.5) the extension $K(x_1)/K$ is totally ramified at π of odd degree $\frac{1}{2}(p-1)$, so adjoining the square root y_1 of the element $y_1^2 = x_1(x_1^2 + a) \in K(x_1)$, which has odd valuation at the (unique, totally ramified) prime above π , strictly doubles the degree; and $\sigma(y_1) = y([\alpha]P_1) = y_1$ forces $\alpha = 1$, so $\rho : \text{Gal}(K(x_1, y_1)/K) \hookrightarrow (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ is injective, hence (by the degree count) an isomorphism onto the cyclic group $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ of order $p-1$. $\square$

Remark 3.7. Theorem 3.6 is identical, word for word in its conclusion, to the $a = 1$ theorem of [6]. This is the precise sense in which the coefficient a plays *no role at all* in the abstract structure of the Galois group: for every nonzero integer a (away from the finitely many primes $\pi \mid 2a$), one gets the same cyclic group $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ of order $N(\pi) - 1$. Anything about E_a that is genuinely sensitive to a must be sought not in the isomorphism type of the group, but in how the concrete field $K(x_1, y_1) \subset \bar{K}$ sits inside $\bar{K}$. This is the subject of the remaining sections.

4 The Quartic Twisting Isomorphism

For $c \in \overline{K}^\times$, the substitution $(x, y) \mapsto (c^2x, c^3y)$ carries E_a isomorphically to E_{a/c^4} :

$$(c^3y)^2 = c^6y^2 = c^6(x^3 + ax) = (c^2x)^3 + (ac^4)(c^2x).$$

Write $[c]$ for this map. Two remarks: (i) for $c = \eta \in \mathbb{Z}[i]^\times$, $[\eta]$ is an automorphism of E_a for every a , and one checks directly that $[\eta]$ commutes with every $[\lambda]$, $\lambda \in \mathbb{Z}[i]$ (both act on the same curve as elements of the commutative ring $\mathbb{Z}[i]$, and $[\eta]$ is literally the endomorphism $[\eta]$ under the identification $\text{Aut}(E_a) = \mathbb{Z}[i]^\times \subset \text{End}(E_a)$, so $[\eta]([\lambda]P) = [\eta\lambda]P = [\lambda]([\eta]P)$); (ii) for general c , $[c]$ is not an endomorphism of any single curve, but it does satisfy $[c] \circ [\lambda]_{E_1} = [\lambda]_{E_{1/c^4}} \circ [c]$ for $\lambda \in \mathbb{Z}[i]$, by the identical computation, since $x([\lambda]P)$ and $y([\lambda]P)$ are homogeneous of the correct weight in (x, y, a) (Lemma 3.3, extended to $\phi_\lambda, \omega_\lambda$ the same way).

Definition 4.1. Fix $t \in \overline{K}^\times$ with $t^4 = a$, and set

$$\phi_t : E_1 \longrightarrow E_a, \quad \phi_t(x, y) := (t^2x, t^3y).$$

Lemma 4.2 (Equivariance). *For every $\lambda \in \mathbb{Z}[i]$, $\phi_t \circ [\lambda]_{E_1} = [\lambda]_{E_a} \circ \phi_t$. In particular $\phi_t(E_1[\pi]) = E_a[\pi]$, and if P_1 generates $E_1[\pi]$ then $\phi_t(P_1)$ generates $E_a[\pi]$.*

Proof. Remark (ii) above with $c = t$, $1/c^4 = 1/a \cdot a = 1$... more directly: by weighted-homogeneity (Lemma 3.3 and its extension to ϕ_n, ω_n), $x([\lambda]P; a)$ evaluated at $(t^2x_0, t^3y_0; a)$ equals $t^2x([\lambda]P; 1)$ evaluated at $(x_0, y_0; 1)$ times the appropriate power of t matching the weight of x , i.e. $x([\lambda](\phi_t(P_0)); a) = t^2x([\lambda]P_0; 1) = x(\phi_t([\lambda]P_0; 1)$ -transported, and similarly in weight 3 for y ; this is precisely the equivariance claimed. Since $[\lambda]P_1 = \mathcal{O}$ iff $[\lambda]\phi_t(P_1) = \phi_t([\lambda]P_1) = \mathcal{O}$ (using ϕ_t injective), ϕ_t carries $E_1[\pi]$ bijectively to $E_a[\pi]$, and it carries a generator to a generator because it is a group isomorphism $E_1(\overline{K}) \rightarrow E_a(\overline{K})$ of $\mathbb{Z}[i]$ -modules. $\square$

The field $K(t) = K(\sqrt[4]{a})$ is Galois over K : since $i \in K$, all four roots of $x^4 - a$ are $t, it, -t, -it$, so $K(t)$ is the splitting field of $x^4 - a$. Write

$$\chi : \text{Gal}(K(t)/K) \longrightarrow \mathbb{Z}[i]^\times, \quad \sigma(t) = \chi(\sigma)t,$$

an injective homomorphism with image the unique subgroup $H \leq \mathbb{Z}[i]^\times$ of order $d := [K(t) : K] \in \{1, 2, 4\}$. Concretely: $d = 1$ iff $a \in (K^\times)^4$; $d = 2$ iff $a \in (K^\times)^2 \setminus (K^\times)^4$; $d = 4$ iff $a \notin (K^\times)^2$. This is Kummer theory for $K \supset \mu_4$; see [3, Ch. VI].

5 The Twisting Theorem

Fix a nonzero π -torsion point $P_1 = (x_1, y_1)$ on E_1 and set $K_1(\pi) := K(x_1, y_1)$, the field of Theorem 3.6 for $a = 1$. By Lemma 4.2, $P_1^{(a)} := \phi_t(P_1) = (t^2x_1, t^3y_1)$ is a nonzero π -torsion point on E_a , so $K_a(\pi) := K(t^2x_1, t^3y_1)$ is the field of Theorem 3.6, with $[K_a(\pi) : K] = N(\pi) - 1$ regardless of a .

Hypothesis 5.1 (Disjointness). $K_1(\pi) \cap K(t) = K$.

This holds whenever the (possible) ramification of $K_1(\pi)/K$, which by [5, Ch. VII] occurs only above $N(\pi)$ and primes of bad reduction of E_1 (here only 2), does not overlap the ramification of $K(t)/K$, which occurs only above primes dividing $2a$; since both fields can in principle ramify at 2 (as $2 \mid \Delta_a$ for every a), Hypothesis 5.1 is not automatic in general, but it is easily checked in specific cases, as we do unconditionally for $N(\pi) = 5$ in §6.

Theorem 5.2 (Twisting theorem). *Let $\pi \nmid 2a$ and assume Hypothesis 5.1. Let $d = [K(\sqrt[4]{a}) : K] \in \{1, 2, 4\}$ and let $H \leq (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ be the (unique) subgroup of order d . Then:*

1. $K_a(\pi)K_1(\pi) = K_1(\pi)(\sqrt[4]{a})$, of degree $d(N(\pi) - 1)$ over K ;
2. $K_a(\pi) \cap K_1(\pi)$ is the unique subfield of $K_1(\pi)$ fixed by H , of degree $(N(\pi) - 1)/d$ over K ;
3. in particular $K_a(\pi) = K_1(\pi)$ if and only if $d = 1$, i.e. $a \in (K^\times)^4$.

Proof. By Hypothesis 5.1, restriction gives an isomorphism

$$\mathrm{Gal}(K_1(\pi)(t)/K) \xrightarrow{\sim} \mathrm{Gal}(K_1(\pi)/K) \times \mathrm{Gal}(K(t)/K) \cong (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times \times H,$$

$\sigma \mapsto (\rho(\sigma), \chi(\sigma))$, where ρ is the identification of Theorem 3.6(3) applied to E_1 . For σ in this group, write $\zeta = \chi(\sigma) \in H \subset \mathbb{Z}[i]^\times$ and $\alpha = \rho(\sigma)$, so $\sigma(x_1, y_1) = ([\alpha]P_1)$. Using $\sigma(t) = \zeta t$ and the equivariance formula $x([\eta]Q; 1) = \eta^2 x(Q; 1)$, $y([\eta]Q; 1) = \eta^3 y(Q; 1)$ for $\eta \in \mathbb{Z}[i]^\times$ (a special case of the weight computation in Lemma 4.2):

$$\sigma(t^2 x_1) = \zeta^2 t^2 \cdot x([\alpha]P_1) = t^2 \cdot x([\zeta\alpha]P_1), \quad \sigma(t^3 y_1) = \zeta^3 t^3 \cdot y([\alpha]P_1) = t^3 \cdot y([\zeta\alpha]P_1).$$

Thus σ acts on $P_1^{(a)} = (t^2 x_1, t^3 y_1)$ exactly as $[\zeta\alpha]$ would, i.e. $\Phi(\sigma) := \chi(\sigma)\rho(\sigma) \in (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ (product in the abelian group $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$) describes the action of σ on $K_a(\pi)$, via the same isomorphism ρ of Theorem 3.6(3) applied to E_a . Since Φ is a homomorphism into an abelian group whose restriction to $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times \times \{1\}$ is already the identity (surjective), Φ is surjective, with $K_a(\pi)$ the fixed field of $\ker \Phi = \{(\zeta^{-1}, \zeta) : \zeta \in H\}$, a subgroup of order d .

This proves $[K_a(\pi) : K] = |(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times \times H|/|\ker \Phi| = d(N(\pi) - 1)/d = N(\pi) - 1$ (consistent with Theorem 3.6), and that $K_a(\pi)$ is a specific index- d subfield of $K_1(\pi)(t)$ distinct from $K_1(\pi)$ (fixed field of $\{1\} \times H$) unless $d = 1$. Composita and intersections now follow from the standard Galois correspondence: $K_a(\pi)K_1(\pi)$ corresponds to $\ker \Phi \cap (\{1\} \times H) = \{(1, 1)\}$, giving all of $K_1(\pi)(t)$, of degree $d(N(\pi) - 1)$, proving (1); and $K_a(\pi) \cap K_1(\pi)$ corresponds to the subgroup generated by $\ker \Phi$ and $\{1\} \times H$, which is $H \times H$ (since $\ker \Phi \cdot (\{1\} \times H) = \{(\zeta^{-1}, \zeta\zeta') : \zeta, \zeta' \in H\} = H \times H$), whose fixed field is the subfield of $K_1(\pi)$ fixed by H (the first factor), of degree $(N(\pi) - 1)/d$, proving (2). Part (3) is the case $d = 1$, $H = \{1\}$. $\square$

Remark 5.3 (Quartic residue interpretation of the twist). When $\pi \nmid 2a$, the extension $K(t)/K$ is unramified at π , and the Frobenius element at π acts on t by $\mathrm{Frob}_\pi(t) = t^{N\pi} = t \cdot t^{p-1} = t \cdot a^{(p-1)/4}$ (using $4 \mid p-1$), so

$$\chi(\mathrm{Frob}_\pi) = a^{(p-1)/4} \bmod \pi = \left(\frac{a}{\pi}\right)_4,$$

the quartic residue symbol of a at π [2, Ch. 9]. Thus the same quantity $a^{(p-1)/4} \bmod \pi$ that appears in Corollary 3.4, controlling irreducibility of ψ_π , also computes the splitting of π in the twisting field $K(\sqrt[4]{a})$: both are shadows of the single fact that $\mathbb{Z}[i]^\times$ embeds in $(\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ as the subgroup of fourth roots of unity, and a acts on the torsion field through its image there.

6 A Complete Worked Example: $N(\pi) = 5$

Let $\pi = 1 + 2i$, so $n = 1, m = 2, N(\pi) = 5$. We first rederive, for general a , the computation of [6, Ch. 4, the worked example with $\pi = 1 + 2i$], which treated only $a = 1$.

Using the standard formulas with $b = 0$: $\phi_1 = x, \psi_1 = 1, \psi_2 = 2y, \phi_2 = x\psi_2^2 - \psi_3\psi_1 = 4x(x^3 + ax) - (3x^4 + 6ax^2 - a^2) = x^4 - 2ax^2 + a^2 = (x^2 - a)^2$. Then

$$\Psi_\pi = \phi_2\psi_1^2 + \phi_1\psi_2^2 = (x^2 - a)^2 + 4x(x^3 + ax) = 5x^4 + 2ax^2 + a^2.$$

One checks directly that

$$\Psi_\pi = ((1 + 2i)x^2 + a)((1 - 2i)x^2 + a),$$

since $(1 + 2i)(1 - 2i) = 5$ and $(1 + 2i) + (1 - 2i) = 2$. By Theorem 3.2, $\psi_\pi(x; a) = (1 + 2i)x^2 + a$. (At $a = 1$ this recovers $\psi_{1+2i}(x) = (1 + 2i)x^2 + 1$ of [6] exactly.) Note $\psi_\pi(0; a) = a$, matching Corollary 3.4 with $(N(\pi) - 1)/4 = 1$ and $u_\pi = 1$.

For $\pi \nmid 2a$ (i.e. $5 \nmid a$, a odd is not required here since $\pi = 1 + 2i \nmid 2$ automatically as $N(\pi) = 5$ is odd – the relevant condition is just $\gcd(a, 5) = 1$), Theorem 3.6 gives $[K(x_1) : K] = 2$ and $[K(x_1, y_1) : K] = 4$, cyclic. Explicitly, $x_1^2 = -a/(1 + 2i) = a \cdot \frac{-1+2i}{5}$; write $c := \frac{-1+2i}{5}$, so $x_1^{(a)} = \sqrt{ac}$ for a fixed square root, and

$$y_1^{(a)2} = x_1^{(a)}(x_1^{(a)2} + a) = \sqrt{ac}(ac + a) = a(c + 1)\sqrt{ac}.$$

6.1 $a = 16$: the trivial twist ($d = 1$)

Here $t = 16^{1/4} = 2 \in K$, so ϕ_t is already defined over K : explicitly $(x, y) \mapsto (4x, 8y)$ carries E_1 isomorphically to E_{16} over K . Concretely $x_1^{(16)} = \sqrt{16c} = 4\sqrt{c} = 4x_1^{(1)}$ and $y_1^{(16)} = \sqrt{16(c+1)\sqrt{16c}}$; substituting $x_1^{(1)2} = c$ one finds $y_1^{(16)2} = 16(c+1) \cdot 4\sqrt{c} = 64 \cdot (c+1)\sqrt{c} = 64y_1^{(1)2}$, so $y_1^{(16)} = 8y_1^{(1)}$. Hence $K_{16}(\pi) = K(4x_1^{(1)}, 8y_1^{(1)}) = K(x_1^{(1)}, y_1^{(1)}) = K_1(\pi)$ *exactly*, as predicted by Theorem 5.2(3): $16 = 2^4 \in (K^\times)^4$, so $d = 1$.

6.2 $a = 4$: the quadratic case ($d = 2$)

Here $4 = 2^2$ is a square in K but not a fourth power ($t^2 = \pm 2$, and neither 2 nor -2 is a square in $\mathbb{Q}(i)$: if $x^2 - y^2 = \pm 2, 2xy = 0$ with $x, y \in \mathbb{Q}$ forces $x = 0$ or $y = 0$, giving $\mp y^2 = 2$ or $x^2 = \pm 2$, none solvable in $\mathbb{Q}$). So $d = 2, t = \sqrt{2}$.

We have $x_1^{(4)} = \sqrt{4c} = 2\sqrt{c} = 2x_1^{(1)} \in K(x_1^{(1)})$, so $K(x_1^{(4)}) = K(x_1^{(1)})$ *identically* – confirming Theorem 5.2(2), since $H = \{\pm 1\}$ is exactly the kernel making $K(x_1)$ the fixed field of $\{\pm 1\} \subset (\mathbb{Z}[i]/\pi\mathbb{Z}[i])^\times$ (Theorem 3.6(1)). At the level of y , however,

$$y_1^{(4)2} = 4(c+1) \cdot 2\sqrt{c} = 8(c+1)\sqrt{c} = 2y_1^{(1)2},$$

so $K_4(\pi) = K(x_1^{(1)})(\sqrt{2}y_1^{(1)})$. This equals $K_1(\pi) = K(x_1^{(1)})(y_1^{(1)})$ if and only if $\sqrt{2} \in K(x_1^{(1)}) = K(\sqrt{c})$, i.e. iff $2/c = 10/(-1+2i) = 2(-1-2i) = -2-4i$ is a square in K . Writing $-2-4i = (m+ni)^2$ with $m, n \in \mathbb{Q}$ forces $mn = -2$ and $m^2 - n^2 = -2$, hence $m^4 + 2m^2 - 4 = 0$, $m^2 = -1 + \sqrt{5} \notin \mathbb{Q}$: impossible. So $\sqrt{2} \notin K(x_1^{(1)})$, and $K_4(\pi) \neq K_1(\pi)$: two *distinct* quadratic extensions of the common field $K(x_1^{(1)})$, with compositum

$$K_4(\pi)K_1(\pi) = K(x_1^{(1)})(y_1^{(1)}, \sqrt{2}),$$

of degree 4 over $K(x_1^{(1)})$, i.e. degree $8 = d \cdot 4$ over K – confirming Theorem 5.2(1). This computation needed no appeal to Hypothesis 5.1: it is a direct, unconditional verification.

6.3 $a = 2$: the maximal twist ($d = 4$)

Here $2 \notin (K^\times)^2$ (shown above), so $d = 4$: $K(\sqrt[4]{2})/K$ has degree 4. At the x -level, $x_1^{(2)} = \sqrt{2c}$, and $K(x_1^{(2)}) = K(x_1^{(1)})$ would require $2c/c = 2$ to be a square in K , which it is not; so $K(x_1^{(2)}) \neq K(x_1^{(1)})$, two *distinct* quadratic extensions of K , hence (quadratic extensions of a field intersect trivially unless equal) $K(x_1^{(2)}) \cap K(x_1^{(1)}) = K$.

Because $K_1(\pi)/K$ and $K_2(\pi)/K$ are each cyclic of degree 4, each has a *unique* quadratic subfield, namely $K(x_1^{(1)})$ and $K(x_1^{(2)})$ respectively (Theorem 3.6(1)). If $K_1(\pi) \cap K_2(\pi)$ properly contained K , being a subfield of the cyclic quartic $K_1(\pi)$ it would have to be $K(x_1^{(1)})$ or all of $K_1(\pi)$; the first forces $K(x_1^{(1)}) = K(x_1^{(2)})$ (false), and the second forces $K_1(\pi) = K_2(\pi)$, hence again $K(x_1^{(1)}) = K(x_1^{(2)})$ (false). So

$$K_1(\pi) \cap K_2(\pi) = K,$$

exactly as predicted by Theorem 5.2(2) with $d = 4 = N(\pi) - 1$, so that $(N(\pi) - 1)/d = 1$. Again, this is an unconditional elementary argument (uniqueness of the quadratic subfield of a cyclic quartic extension), independent of Hypothesis 5.1, which for this particular π we have therefore verified rather than assumed.

7 Concluding Remarks

The upshot is a clean dichotomy. As an *abstract group*, $\text{Gal}(K(x_1, y_1)/K)$ for the family $E_a : y^2 = x^3 + ax$ is completely insensitive to a : it is cyclic of order $N(\pi) - 1$ for every $a \in \mathbb{Z} \setminus \{0\}$ and every split $\pi \nmid 2a$, by Theorem 3.6, whose proof is the $a = 1$ proof of [6] with one corrected lemma (Corollary 3.4) about the constant term. As a *concrete subfield of $\overline{K}$* , however, $K(x_1, y_1)$ moves with a in a way governed entirely by the class of a in $K^\times/(K^\times)^4$ – a purely global, π -independent invariant – via the twisting isomorphism of §4

and the quartic residue symbol $\left(\frac{a}{\pi}\right)_4$ of §5. The worked example of §6 shows all three cases $d = 1, 2, 4$ occurring already at the smallest split prime, and shows that at least there, the field-theoretic consequences of Theorem 5.2 can be verified with no hypothesis at all, by the elementary observation that a cyclic extension of prime-power degree has a unique subfield of each degree dividing it.

It would be of interest to remove Hypothesis 5.1 in general, presumably by a more careful local analysis at the shared prime 2 (which divides $\Delta_a = -64a^3$ for every a , and may divide the conductor of $K(\sqrt[4]{a})/K$ as well), and to determine whether the quartic residue symbol $\left(\frac{a}{\pi}\right)_4$, beyond controlling the splitting of π in $K(\sqrt[4]{a})/K$ as in the remark closing §5, also determines finer invariants of $K_a(\pi)$ itself (for instance, its ramification at primes dividing a), in the spirit of the general theory of complex multiplication and ring class fields [1].

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